

Learner Guidance

Using Index Laws · How can patterns help us simplify powers without expanding every factor?

You can pause and return.

You do not need to complete everything in one sitting. Using support, including a Reconnection Route, is part of learning.

What are we exploring?

You will connect index laws to repeated multiplication. The aim is to understand why each exponent changes, rather than memorising three disconnected rules.

Your Stepping Stones

- recognise the base and exponent in a power;
- multiply powers with the same base by adding exponents;
- divide powers with the same base by subtracting exponents;
- raise a power to another power by multiplying exponents;
- use the zero-index pattern when a non-zero power is divided by itself;
- explain a law using repeated factors.

Before you begin

- You may use the in-lesson Scratchpad, paper, a physical whiteboard or another recording method.
- You can expand a small power when you want to see the pattern.
- Choose a quieter space, reduce screen brightness or take a movement break if that helps.

Reconnection Routes

The Reconnection Routes button is near the top of the lesson. Choose a short recap only if multiplication, powers, base/exponent language or operation symbols are blocking the current task. Every route returns to Using Index Laws.

A possible way through the lesson

1. Connection: become an Index Law Detective and repair an incorrect claim.
2. Movement: use the Exponent Pattern Machine and predict the resulting exponent.
3. Reflection: choose the law before calculating and explain why it fits.
4. Creativity: build three expressions that simplify to the same target power.
5. Rest: retrieve same base $\rightarrow$ operation $\rightarrow$ exponent rule.
6. Nutrition: notice how compact notation nourishes structure, reasoning, efficiency and precision.
7. Final Task: design and explain an index-law pathway.

Timings are invitations, not deadlines.

You may stop after a meaningful piece of thinking and continue later.

Using the Exponent Pattern Machine

8. Choose a positive whole-number base.
9. Choose multiply powers, divide powers or power of a power.
10. Enter two exponents from 0 to 12.
11. Predict the resulting exponent.
12. Check your prediction.
13. Reveal the repeated-factor explanation.

For division, a smaller first exponent can produce a negative exponent. Negative indices are optional stretch in this lesson; the core pathway uses non-negative resulting exponents.

Practice Companion and Scratchpad

Practice Companion

The six questions cover multiplication, division, power of a power, a missing exponent, misconception repair and the zero-index pattern. Enter the resulting exponent only.

Universal worked-solution release rule

For every question independently: Socratic prompt used + Hint used + two non-empty answers entered and checked = Worked solution released. The conditions begin again for each question.

The button tells you what remains. A Supporting Adult should not provide the complete worked solution while it is locked.

Index-law Organiser Scratchpad

- Original expression: copy the power expression.
- Same base?: check before selecting a law.
- Operation: multiplication, division or power of a power.
- Exponent rule: add, subtract or multiply.
- Simplified power: keep the base and write the new exponent.

When you feel stuck

- Expand a small example as repeated multiplication.
- Circle the base and underline each exponent.
- Use one Reconnection Route, then return.
- Say the operation aloud before choosing a rule.
- Pause and come back.

Showing what you understand

You may show understanding through typed expressions, spoken explanation, annotations, a Scratchpad image, paper working or a combination.

A useful explanation includes

- the original expression;
- confirmation that the bases match;
- the operation being performed;
- the exponent calculation;
- the simplified power;
- one repeated-factor explanation.

Short self-review

- | | | | |
|----------------------------------|---------------------------------------|---|---|
| ☐ Not yet | ☐ With support | ☐ Secure for now | I can identify the base and exponent. |
| ☐ Not yet | ☐ With support | ☐ Secure for now | I can choose between adding, subtracting and multiplying exponents. |
| ☐ Not yet | ☐ With support | ☐ Secure for now | I keep the base unchanged when the law applies. |
| ☐ Not yet | ☐ With support | ☐ Secure for now | I can explain one law using repeated factors. |

A valid finishing point

Stop when you have made one accurate index-law choice and can name your next connection.
Incomplete work is information, not failure.